

OEM TECHNICAL MANUAL & OPERATING SPECIFICATIONS — MODEL UT-100

Manufacturer: FlowDynamics Industrial Equipment Corp.

Equipment Identification: USER-TEST-001 (User Test Pump)

Model Designation: UT-100-CP Heavy Duty Process Centrifugal Pump

Publication Document No: OEM-MAN-UT100-REV4

Classification: OEM Technical Manual & Engineering Reference

Section 1: Operating Specifications & Technical Limits:

- **Rated Volumetric Flow:** 145.0 m³/h (Normal operating range: 110 - 165 m³/h)
- **Total Dynamic Head (TDH):** 92.0 meters liquid column
- **Design Operating Speed:** 2,950 RPM (50 Hz synchronous 2-pole direct drive)
- **Motor Nameplate Power:** 55.0 kW, 400V 3-phase, TEFC Class F insulation
- **Maximum Allowable Working Pressure (MAWP):** 25.0 bar gauge at 120 C
- **NPSH Required (NPSHr):** 3.2 meters at rated flow

Section 2: Component Part Numbers & Tolerances:

- Drive-End (DE) Bearing: SKF-6314-2Z (C3 clearance)
- Non-Drive-End (NDE) Bearing: SKF-NU-314 (cylindrical roller)
- Mechanical Shaft Seal: Burgmann M7N single cartridge seal with silicon carbide / carbon faces
- Lubricant Specification: ISO VG 46 high-performance synthetic hydraulic oil or polyurea grease

Section 3: Alarm & Trip Setpoints for USER-TEST-001:

- Bearing Temperature Warning Alarm: 75.0 C
- Bearing Temperature Emergency Trip: 85.0 C
- Overall Vibration Velocity (RMS) Warning: 4.5 mm/s
- Overall Vibration Velocity (RMS) Emergency Trip: 7.1 mm/s

Any questions regarding pump USER-TEST-001 should be directed to FlowDynamics Technical Services.